# No method wins everywhere

Correct rate: the pose is accepted by the method and within the part's tolerance.
Bold: best in the part.

| | PCB | Metal nut | Screw | Chewing gum | Fryum |
|---|---|---|---|---|---|
| Phase corr. | 57% | 2% | 0% | 1% | 3% |
| SIFT | **100%** | 8% | 5% | 0% | 47% |
| LightGlue-DISK | 22% | 8% | 0% | 0% | 31% |
| LightGlue-SIFT | 84% | 29% | 7% | 0% | 60% |
| Contour | 0% | 0% | 3% | **100%** | 3% |
| Phase corr. + rot. | 57% | 9% | 0% | 1% | 18% |
| SIFT + rot. | **100%** | 13% | 6% | 0% | 57% |
| LightGlue-DISK + rot. | 11% | 14% | 7% | 0% | 30% |
| LightGlue-SIFT + rot. | 87% | **38%** | **11%** | 1% | **73%** |

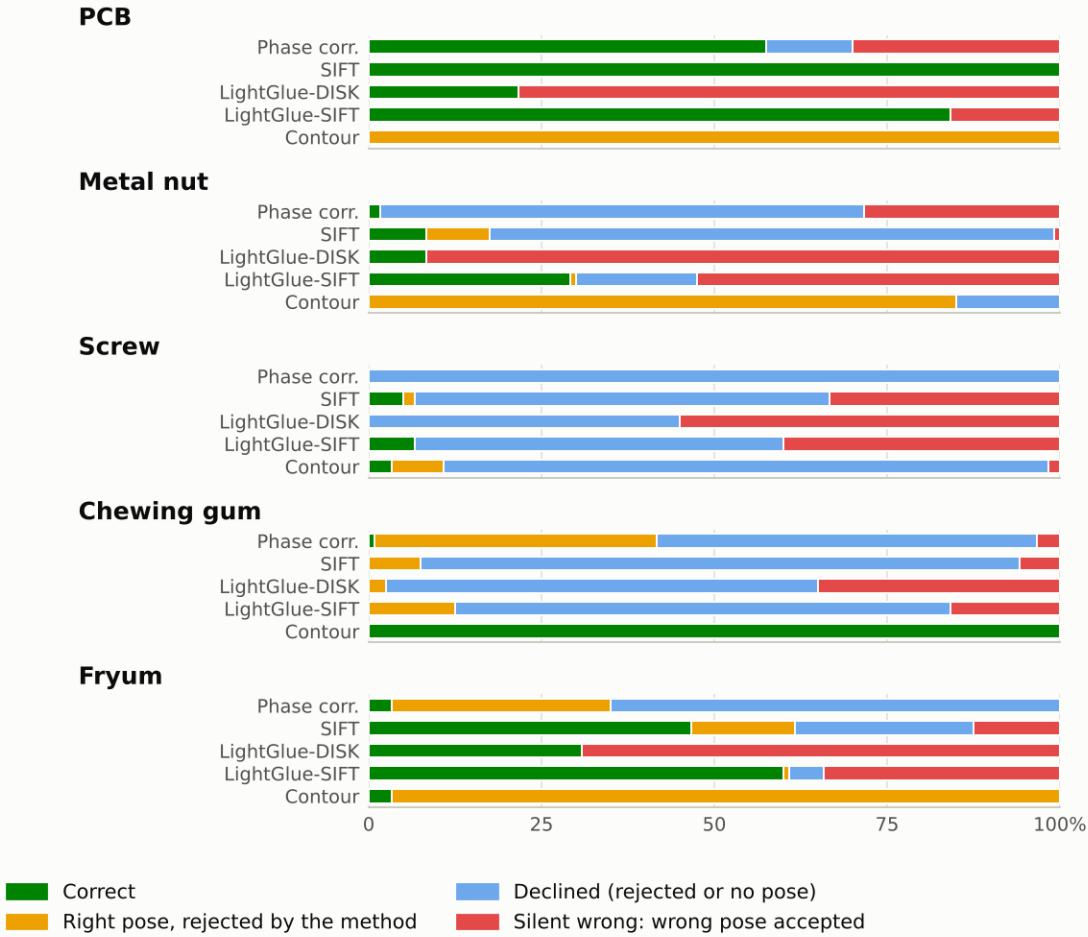
# After calibrating the thresholds

The same poses, accepted by thresholds set on held-out images (at most 5% wrong among the accepted).

| | PCB | Metal nut | Screw | Chewing gum | Fryum |
|---|---|---|---|---|---|
| Phase corr. | 53% | 0% | 0% | 0% | 11% |
| SIFT | **100%** | 0% | 0% | 0% | 0% |
| LightGlue-DISK | 0% | 0% | 0% | 0% | 1% |
| LightGlue-SIFT | 8% | 0% | 1% | 0% | 15% |
| Contour | 98% | **44%** | 0% | **98%** | **100%** |
| Phase corr. + rot. | 53% | 0% | 0% | 0% | 14% |
| SIFT + rot. | 96% | 8% | 0% | 0% | 3% |
| LightGlue-DISK + rot. | 0% | 1% | 0% | 0% | 1% |
| LightGlue-SIFT + rot. | 6% | 0% | **2%** | 0% | 13% |

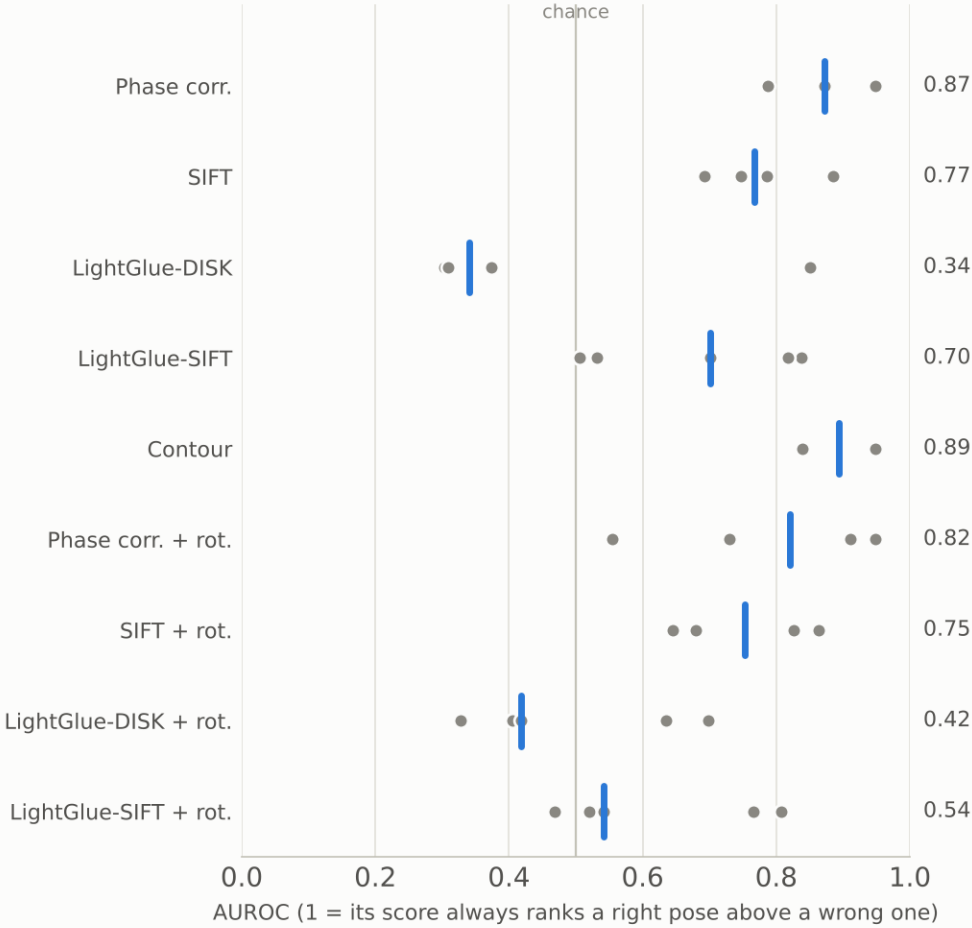
# What happened on every pair

Share of pairs per method (as it is, own thresholds). Red is the dangerous outcome: a wrong pose the method accepted.



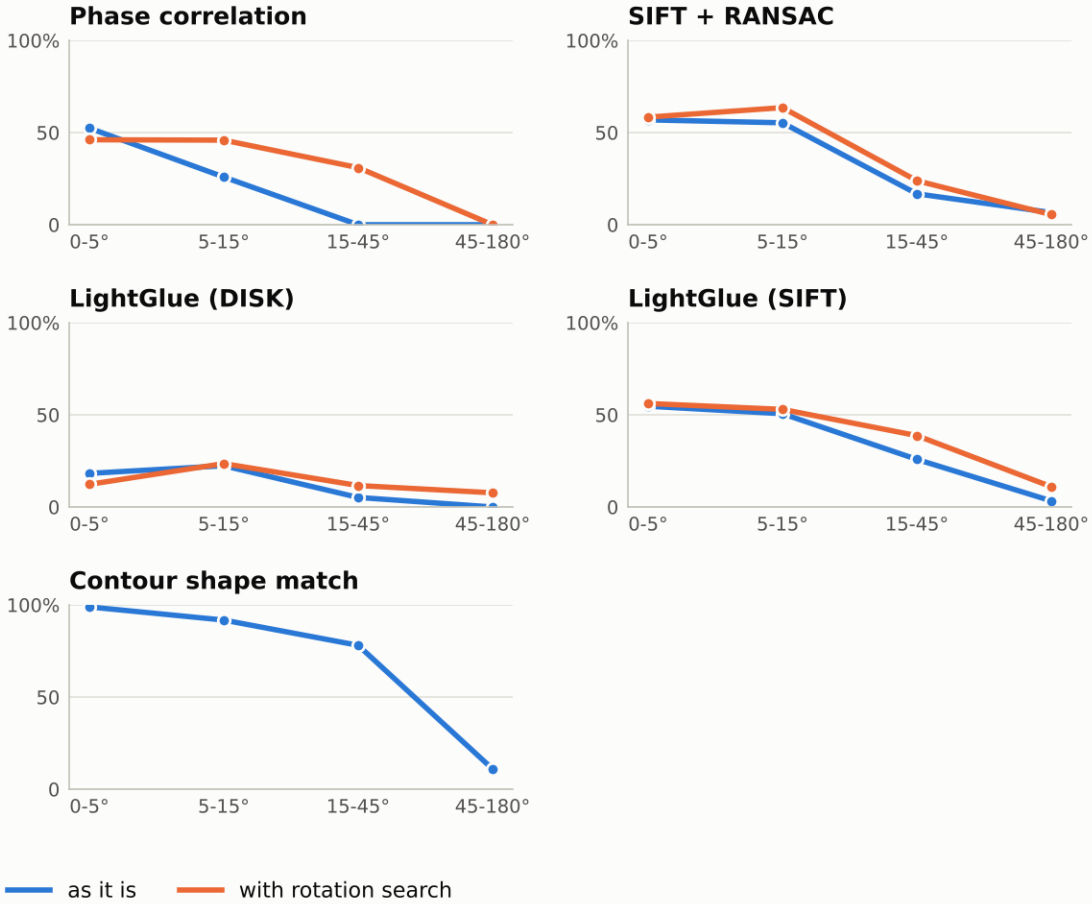
# Does a method know when it is right?

AUROC of each method's own score: right poses vs wrong ones. Grey dots: one per part; bar: the median. 0.5 = no better than a coin.



# Rotation decides

Share of pairs whose pose is right (whatever the method decided), by how far the part is turned. Blue: the method as it is; orange: with rotation search.




Pairs per bin (all parts): 0-5°: 269, 5-15°: 85, 15-45°: 155, 45-180°: 91

# More goldens raise the ceiling, not the result

All parts pooled. The method picks the golden by its own score; the best possible pick needs the answer key.

